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Unit-1: Blockchain Fundamentals -

① Explain any three important Properties of blockchain and their significance.

→ Blockchain technology has several Properties

ⓐ Decentralization: Blockchain Operates through a Peer to Peer network in which control is distributed among multiple nodes rather than a single central authority.

ⓑ Immutability → once a block is added to the blockchain modifying its contents becomes extremely difficult.

ⓒ Transparency → Transactions recorded on a Public blockchain can be viewed and independently verified by network participants. This improves accountability and trust.

These Properties make blockchain suitable for crypto currencies, supply-chain tracking and record management.

② Differentiate between a centralized database and a blockchain with respect to data management, control, and security.

→ A centralized database is managed by a central authority, whereas a blockchain distributes ledger ~~maintain~~ maintenance across participating nodes.

ⓐ Data management → A centralized database stores and manages records under a central system. A blockchain maintain replicated ledger copies across participating nodes.

(b) Control → In a centralized database, an administrator generally controls access, updates and deletion. In a blockchain, transaction acceptance follows predefined rules and a consensus mechanism.

(c) Security → A centralized database may become a single point of failure if its central server is compromised. Blockchain uses cryptographic hashes, digital signatures, and distributed validation to protect data integrity.

(d) Modification → centralized records can generally be updated by authorized administrators. Confirmed blockchain records are designed to be tamper-resistant.

(3) Explain the structure of a blockchain block with a neat diagram, showing the block header, transaction data and previous block hash.

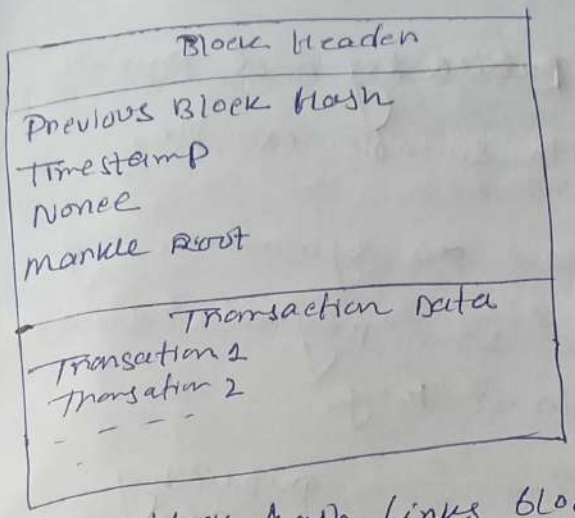
→ A Blockchain block is a container that stores transaction information and metadata. It consists of 2 major parts: the block header and transaction data.

The block header generally contains the following fields:

1. Previous block hash: cryptographically links the current block to its predecessor.
2. Timestamp: Records the approximate time associated with block creation.
3. Nonce: A variable used in Proof-of-work mining.

* Merkle Root: Represents the hashes of all transactions in a block.

The transaction data contains the transactions included in the block



The Previous block hash links blocks together, forming a chronological tamper-resistant chain.

④ What is block mining? Explain the major steps involved in mining a block in a Proof of work Blockchain?

→ Block mining is the process of creating a new block and adding it to a Proof-of work Blockchain.

The major steps are.

(a) Collect transactions: miner collect pending transactions from the network

(b) Validate transactions → They check transaction authenticity and compliance with blockchain rules.

(c) Solve the PUZZLE → miners repeatedly calculate block hashes while changing the nonce.

(d) Broadcast the Block: The successful miner broadcasts the proposed block to the network.

(e) Verify and add → Others nodes in dependantly (4) Verify the block. if valid, they accept it accordely to the blockchain's consensus rules.

(6) Different between public key and private key cryptography with suitable example

→ Private key is called as symmetric key cryptography. Use the same key and public key uses the different key

Private key cryptography	public key cryptography
(1) one shared key	(a) public and private key ~~shared~~
(2) use the secret key for encryption	(b) use the receiver public key.
(3) use the same key for decryption.	(c) use the receiver private key
(4) ~~Per~~formance faster	(d) Generay more slow.
(5) Ex → AES, DES	(e) RSA.

(7) Define a cryptographic hash function and explain any 4 important properties required for blockchain security?

→ A cryptographic hash function is a mathematical algo. that converts an input of arbitrary length into a fixed length output called a hash value or message digest. Blockchain uses hashing to link blocks and verify data integrity.

4 important properties are,

(a) Deterministic → The same input always produces the same hash.

(b) Preimage resistance → Given a hash value finding an input that produces it should be infesible

(c) second preimage resistance → Given a hash one input to find a different input with the same hash

A small change in input should produce a different output known as the avalanche effect.

(8) Explain the construction of a merkle tree for 4 transactions with a neat diagram. What is the significance of the merkle root?

→ A merkle tree is a tree like structure used in blockchain to store and verify transaction hashes efficiently. It is constructed by combining transaction hashes until a single hash, called the merkle root.

Suppose there are 4 transactions: T_1, T_2, T_3 and T_4

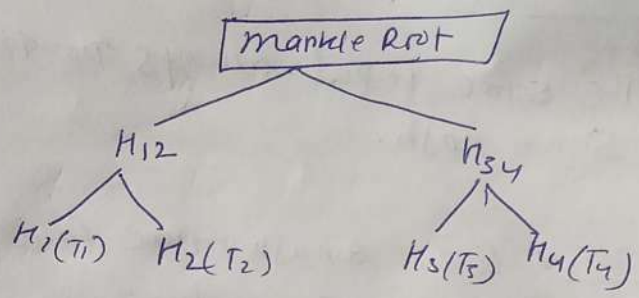
Steps →

(a) First, calculate the hash of each transaction $H(T1)$, $H(T3)$, $H(T4)$

(b) Combine the 1st 2 hashes and calculate a new hash: $H12$

(c) Combine the next 2 hashes and calculate another hash: $H34$

(d) Combine $H12$ and $H34$ to obtain the final merkle root.



(19) In an RSA system, let $P=5$, $q=11$ and the Public exponent $e=3$. calculate n, $\phi(n)$ and the private exponent d. Show the necessary steps.

→ RSA is a public key cryptographic algo that uses 2 keys a public key and a private key.

$P=5$

$q=11$

$e=3$

Step 1: Calculate $n = P*q$

$$n = 5 * 11 = 55$$

Step 2 → Calculate Euler's totient function

$$\phi(n) = (P-1)(q-1)$$

$$= (5-1)(11-1)$$

$$= 4*10 = 40$$

Step-3: Calculate private exponent d

$(e \times d) \bmod \phi(n) = 1$

$\therefore (3 * d) \bmod 40 = 1$

$d = 27$

$3 * 27 = 81$

$81 \bmod 40 = 1$

Therefore $d = 27$

∴ Public key = $(n, e) = (55, 3)$

Private key = $(n, d) = (55, 27)$

(12) Explain the process of creating and verifying a digital signature using public key cryptography.

→ A digital signature is a cryptographic technique used to verify the identity of the sender and ensure that a message has not been changed during transmission.

It provides authentication, integrity and supports non-repudiation.

* Creating a digital signature:

1. The sender creates a message.
2. A hash function generates a message digest.
3. The sender signs the digest using their private key.
4. The message and digital signature are sent to the receiver.

* Verifying a digital signature →

1. The receiver receives the message and signature
2. The receiver calculates a new hash of the received message.
3. The receiver verifies the signature using the sender's public key.

Verification succeeds, the signature is valid and the message has not been modified.

Senden side.

original message
↓
Hash function
↓
Sign digest using Private key
↓
Digital signature

Receiver side.

Verify using Senden's Public key
↓
Valid sign if verification sueeeds.

(13) Write down the Properties of message digest in details?

→ A message diggest is a fixed length value generated from a message using hash function. It acts like a digital ~~fi~~ fingerprint and helps check whether the original message has been changed.

*important Properties of message diagest are →

(1) ~~Easy computation~~ →

(1) Fixed length outfut: A message of any size produces a fixed length digest. For example SHA-256 ~~produ~~ Produces 256 bits.

(2) Easy computation → The digest should be calculated quicky for any given message.

(3) Collision resistance → it should be diffcult to find 2 ~~diff~~ different message that Produces same digest.

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④ Message Avalanche check → Even a small change in the message should produce a significantly different digest.

Unit-III Distributed system and cryptocurrency.

(14) Define a distributed system. Explain how blockchain technology uses the principles of distributed system.

→ A distributed system is a collection of multiple ~~indep~~ independent called nodes that communicate with each other and work together to perform a common task.

*Blockchain uses the Principles of distributed system in the following ways:-

① Decentralization → Data is maintained by multiple computer instead of one central authority.

② Peer-to Peer communication → Nodes communicate directly and share transactions and blocks.

③ Distributed ledger → Multiple nodes maintain copies of the blockchain ledger

④ Consensus mechanism → Nodes agree on which transactions and blocks are valid.

⑤ Faut tolerance → The network can continue working even if some nodes fail.

⑥ Security: cryptographic hashing and digital signatures Project transation integrity.

Ex → Bitcoin uses a distributed network where different computers maintain and verify the same transaction ledger without a central ~~bank~~ bank.

(15) Discuss the major design goals of crypto-currencies. (10)

→ Cryptocurrencies are digital currencies that use cryptography and blockchain technology to transfer money securely without depending on a central financial authority.

* The major design goals are →

(1) Decentralization → No single person or organization controls the entire network.

(2) Security → Cryptography protects transactions from unauthorized access and manipulation.

(3) Privacy → Users can transact using wallet addresses without directly displaying their real identities.

(4) Integrity → Transaction records are protected against unauthorized modification.

Ex → Bitcoin allows users to send and receive digital currency without requiring a traditional bank to process every transaction.

(17) Differentiate between desktop wallets, mobile application based wallets and browser based wallets.

→ A cryptocurrency wallet is a software application used to store and manage private keys and public keys. It helps users send, receive and manage cryptocurrencies

Feature	Desktop wallet	Mobile Wallet	Browser-based wallet
Platform	Computer or laptop	Smartphone	Web Browser or extension.
installation	installed on computer	installed as mobile app	Accessed through browser.
Convenience	Suitable for desktop users	Easy to use anywhere	Easy access to web applications.
Security risk	Computer malware	malware	Phishing attacks.
EX→	Electrum	Trust wallet	metamask

(18) Explain the role of nodes in a peer-to peer blockchain network and how they maintain a shared ledger.

→ A node is a computer connected to a blockchain network that participates in communicating storing or validating blockchain information.

Nodes play an important role in maintaining a shared and secure blockchain ledger.

*main roles of nodes →

(1) Transaction verification.

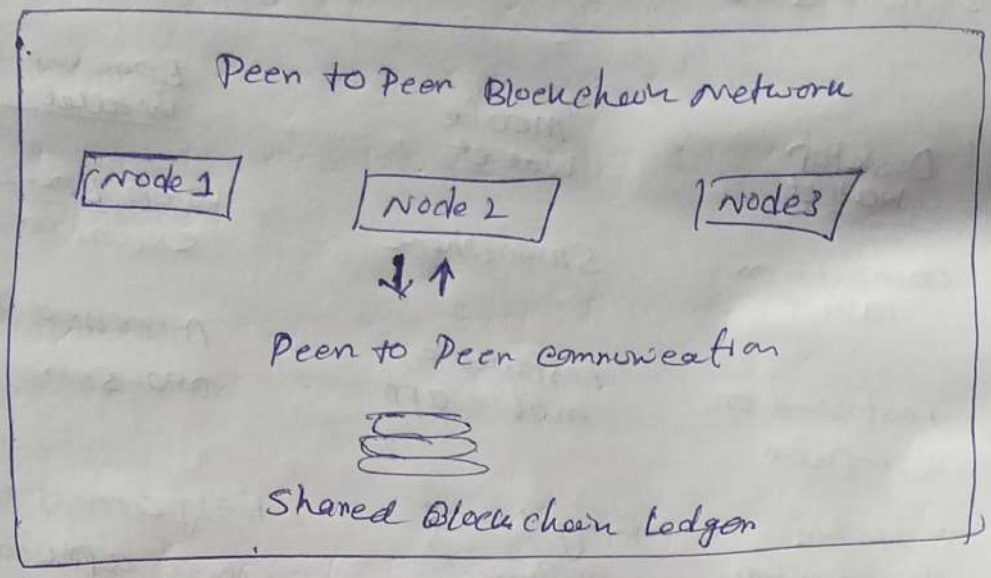
(2) Transaction Broadcasting.

③ Block Verification

④ Ledger maintance

⑤ Data Sharing

⑥ ~~Consus~~ Consensus Participation.



Unit-IV: Consensus, Ethereum and Hyperledger

(21) Explain the Byzantine Generals Problem and its relevance to blockchain consensus mechanisms.

→ The Byzantine Generals Problem is a problem in distributed systems where different computers must agree on a common decision even when some computers are faulty or malicious.

Ex→ Imagine 4 ~~any~~ army generals surrounding an ~~em~~ eny city. They must decide whether to attack ~~or~~ t They communicate through message, but some generals may send false information

⑧ Relevance to blockchain →

(i) Agreement → All participating nodes must agree on the valid transactions.

(ii) Fault tolerance → The network should continue working even if some nodes are faulty.

(iii) Reliability → It helps maintain a consistent ledger.

The Byzantine Generals Problem explains how blockchain networks achieve agreement without trusting every participant.

⑨ Explain Nakamoto consensus and role of Proof of work in selecting the accepted blockchain.

→ Nakamoto consensus is the consensus mechanism introduced by Bitcoin to help network participants agree on a common blockchain without a central authority.

It combines proof of work (PoW) mining and the selection of the chain with the greatest cumulative Proof of work.

* Working Process →

(1) Users create and broadcast transactions to the network.
(2) miners collect valid transactions into a candidate block.
(3) Miners perform proof of work by solving a computational puzzle.
(4) The successful miner broadcasts the new block.
(5) Other nodes verify the block and accept it if it follows the network rules.

(5) When competing valid chains exist, nodes follow the chain with the greatest cumulative Proof of work. (4)

(6) Role of Proof of work →

(1) makes block creation computationally difficult.

(2) Prevents easy manipulation of transaction history.

(3) Helps to resolve competing blockchain versions.

(7) Nakamoto consensus helps Bitcoin maintain a secure and shared ledger without a central authority.

(8) Compare Proof of ~~State~~ stake, Proof of Activity, Proof of Burn, Proof of Authority, Proof of Elapsed Time, and Proof of Performance.

→ There are different consensus mechanisms used by blockchain networks to validate transactions and crete new blocks.

(1) Proof of stake (PoS) → Validators are selected based on cryptocurrency they lock as stake.

(2) Proof of Activity (PoA) → combines Proof of work mining with Proof of stake validation

(3) Proof of Burn (PoB) → Participants Permanently destory coins to demonstrate commitment to the network.

(iv) Proof of Authority (PoA) → Approved and trusted validators create and validate blocks.

(v) Proof of Elapsed Time (PoET) → Nodes wait for randomly assigned time periods, the eligible shortest wait gets the opportunity to ~~create~~ create a block

(vi) Proof of importance (PoI) → selects participants based on an importance score that can consider balance and network activity

(26) What is smart contract? Explain how smart contracts are developed using solidity and executed on ~~Ethereum~~ Ethereum.

→ A smart contract is a self executing computer program stored on a blockchain. It automatically performs actions when predefined conditions are satisfied, without requiring an intermediary.

Solidity is a programming language used to develop smart contracts on Ethereum.

Steps

1. writing code
2. compilation
3. deployment
4. Execution
5. verification
6. Gas fees.

Ex → A smart contract can automatically transfer money when a buyer receives a product (16)

Smart contracts automate agreements and support decentralized application on Ethereum

Unit-V : Blockchain Applications →

(27) How is blockchain applied in land registry records? Explain in detail

→ Blockchain technology can be used to store and manage land ownership records securely. It creates a transparent and tamper-evident digital records of lands transactions.

Working Process →

(i) Digital registration → Land details and ownership information are recorded digitally.

(ii) Transaction verification → Government authorities verify the ownership and property documents.

(iii) Transparency → Authorized users can check the history of land ownership.

(iv) Security → cryptographic hashing makes unauthorized change to records detectable

Ex- if a person sells a piece of land, the verified ownership transfer can be recorded on a government blockchain network

(23) Discuss four blockchain governmental applications of blockchain technology. (12)

→ Blockchain technology can improve government services by providing transparency, security and efficient record management.

4 important governmental application →

(1) Land registration →

(a) Stores Land ownership records securely

(b) Reduces Property fraud and disputes.

(2) Digital identity →

(a) Helps maintain secure digital identity records.

(b) Allows authorized verification of citizen's identities.

(3) voting system →

(a) Records votes secury and transparenty.

(b) makes unauthorized chages detectable

(4) Public fund management →

(a) Records government spendy and transactions

(b) Helps reduce ~~de~~ corruption and missue of ~~fund~~ funds.

Blockchain can make government services more transparent secure and efficient

(20) Explain how blockchain can be used for supply-chain tracking and product provenance. Discuss its benefits and challenges.

→ A supply chain is the process of moving products from manufactures to customers. Product Provenance means tracking the origin and history of a product.

Blockchain helps record every stage of a product's journey in a secure and transparent manner.

Working Process →

(i) Manufacturing → Product details are recorded when the product is created

(ii) Transportation → Shipping and delivery details are added to the blockchain.

(iii) Distribution → Retailers update the product's movement

Benefits →

(a) Transparency: Participants can view product history

(b) Authenticity → Helps identify counterfeit products.

(c) Trust → improves confidence among customers and suppliers.

(x) Challenges →

(a) High implementation cost
(b) Incorrect information entered into the system.
(c) Integration with existing systems.
(d) Privacy and scalability issues.

Blockchain improves supply chain transparency, Product traceability and trust, but accurate data entry and Proper implementation are essential.

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